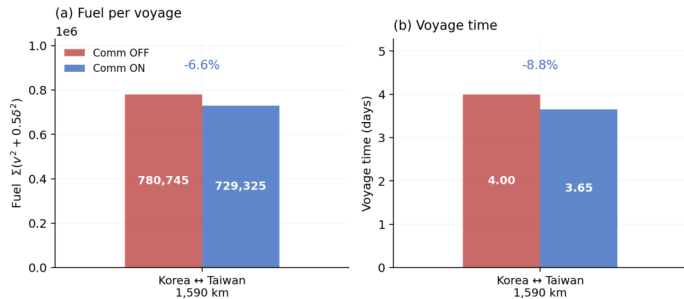


(a) Real-coastline global A* route (Busan–Taiwan)

Korea ↔ Taiwan corridor · 10,000 vessels · bidirectional



(b)

Simulator route example

One obstructed spawn–goal pair:
direct line · A* waypoints · obstacles
· inflated boundary · goal

pending re-evaluation

(c)

Collision rate

4 conditions × {all 320, blocked 148}
Direct/A* × Comm OFF/ON

pending re-evaluation

(d)

Arrival rate

4 conditions × {all 320, blocked 148}
Direct/A* × Comm OFF/ON

pending re-evaluation

Panel (a) is a coastline occupancy map built from Natural Earth polygons — it is not an AIS traffic digital twin.
Panels (b)–(d) await the rerun specified in the reorganization plan: proposed checkpoint (qd_MOE_SE), ring = 0.7, direct vs A*, communication OFF vs ON, reporting all 320 spawn–goal pairs and the 148 blocked pairs separately. 172 / 320 pairs have a clear direct line, so A* and direct coincide there.
This is zero-shot waypoint integration: the policy was trained on direct goals and was not retrained for waypoint following.